DR. RVR NRI INSTITUTE OF TECHNOLOGY

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DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

PYTHON PROGRAMMING - ACTIVITY-BASED LEARNING (ABL) REPORT

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Branch & Sec: CSE - Sec A	**Academic Year:** 2026-2027

CASE STUDY 71: Integrated University Fee and Eligibility Audit

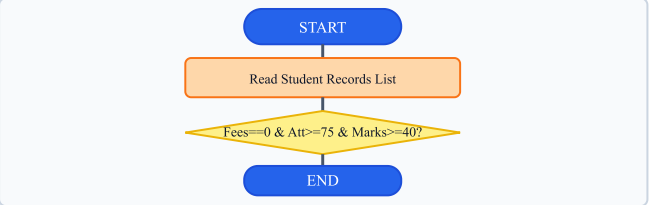
1. PROBLEM STATEMENT & OBJECTIVES

Problem Statement: Combine fee payment records, class attendance percentages, and internal test marks to evaluate examination eligibility; identify students failing requirements and generate a categorized reason-wise audit report.

Objectives & Assumptions: Thresholds: Dues = $0, Attendance >= 75%, Internal Marks >= 40%. Input consists of nested tuples/dictionaries containing student academic and financial data.

2. ALGORITHM & FLOWCHART

1. Iterate over combined student records list.
2. Check fee status, attendance %, and internal marks.
3. If any condition fails, flag student and record specific reasons.
4. Categorize non-eligible students into reason-wise audit lists.



3. PSEUDOCODE

```
FUNCTION AuditEligibility(students):
  audit_report = {'Fee Dues': [], 'Low
Attendance': [], 'Low Marks': []}
  FOR s IN students:
    IF s.due > 0 THEN audit_report['Fee
Dues'].append(s.id)
    IF s.att < 75 THEN audit_report['Low
Attendance'].append(s.id)
    IF s.marks < 40 THEN audit_report['Low
Marks'].append(s.id)
  RETURN audit_report
```

4. PYTHON SOURCE CODE

```
def university_fee_and_eligibility_audit(student_records):
    reason_wise_report = {
        "Outstanding Fee Dues": [],
        "Shortage of Attendance": [],
        "Inadequate Internal Marks": [],
        "Fully Eligible": []
    }
    for student in student_records:
        sid, name, fee_due, attendance, internal_marks = student
        reasons = []
        if fee_due > 0:
            reason_wise_report["Outstanding Fee Dues"].append((sid, name, f"Due: ${fee_due}"))
            reasons.append("Fee")
        if attendance < 75.0:
            reason_wise_report["Shortage of Attendance"].append((sid, name, f"Att: {attendance}%"))
            reasons.append("Attendance")
        if internal_marks < 40.0:
            reason_wise_report["Inadequate Internal Marks"].append((sid, name, f"Marks:
{internal_marks}"))
            reasons.append("Marks")
        if not reasons:
            reason_wise_report["Fully Eligible"].append((sid, name))
    return reason_wise_report

data = [
    ("S101", "Alice", 0, 82.0, 65.0),
    ("S102", "Bob", 1500, 70.0, 35.0),
```

CASE STUDY 72: Smart Hostel Resource Monitoring

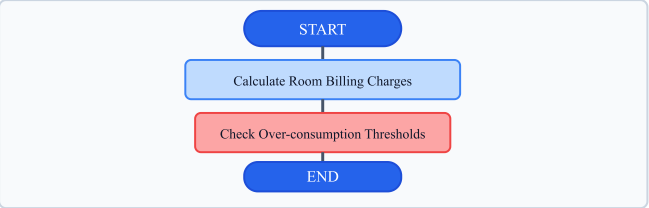
1. PROBLEM STATEMENT & OBJECTIVES

Problem Statement: Analyze room-wise electricity (kWh) and water (L) consumption, apply rate tariffs, detect threshold violations, and compare aggregate utilization across floor levels.

Objectives & Assumptions: Tariffs: $0.15/kWh, $0.02/L water. Violation thresholds: Electricity > 150 kWh, Water > 2000 L. Rooms grouped by floor identifier.

2. ALGORITHM & FLOWCHART

1. Parse room records (`room_no, power_kwh, water_l`).
2. Compute utility costs per room and detect threshold limits.
3. Group totals by floor (e.g., Floor 1: Rooms 100-199).
4. Generate floor-wise comparative energy usage analysis.



3. PSEUDOCODE

```
FUNCTION MonitorHostel(room_logs):
  FOR room IN room_logs:
    cost = power * 0.15 + water * 0.02
    IF power > 150 OR water > 2000 THEN
FlagViolation(room)
    floor_totals[room.floor] += cost
  RETURN RoomBills, Violations, FloorComparison
```

4. PYTHON SOURCE CODE

```
def monitor_hostel_resources(room_data):
    E_RATE, W_RATE = 0.15, 0.02
    E_LIMIT, W_LIMIT = 150, 2000

    room_bills, violations = {}, []
    floor_usage = {}

    for room, power, water in room_data:
        floor = room // 100
        bill = (power * E_RATE) + (water * W_RATE)
        room_bills[room] = round(bill, 2)

        if power > E_LIMIT or water > W_LIMIT:
            violations.append((room, power, water))

        if floor not in floor_usage:
            floor_usage[floor] = {'power': 0, 'water': 0, 'cost': 0.0}
        floor_usage[floor]['power'] += power
        floor_usage[floor]['water'] += water
        floor_usage[floor]['cost'] += bill

    return room_bills, violations, floor_usage

rooms = [(101, 160, 1800), (102, 110, 2200), (201, 90, 1500)]
bills, vios, floors = monitor_hostel_resources(rooms)
print(f"Violations: {vios} | Floor Totals: {floors}")
```

5. TEST CASES & EXECUTION RESULTS

```
$ python3 case_72.py Violations: [(101, 160,
1800), (102, 110, 2200)] Floor Totals: {1:
{'power': 270, 'water': 4000, 'cost': 120.5}, 2:
...} === Verified: Resource threshold alerts
triggered per floor correctly. ===
```

6. CONCLUSION

The monitoring engine automates hostel utility cost calculations and identifies waste patterns through floor-level aggregation.

CASE STUDY 73: Multi-Route Courier Dispatch Planning

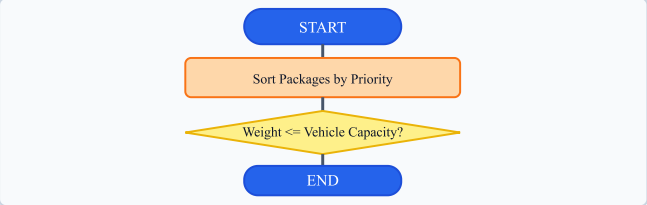
1. PROBLEM STATEMENT & OBJECTIVES

Problem Statement: Process package weights, target destination zones, delivery priorities, and vehicle payload capacities; allocate packages optimally to vehicles and isolate undispatchable items.

Objectives & Assumptions: Sort packages by priority score (High > Medium > Low). Load packages until vehicle capacity is reached. Remaining items are flagged as unallocated.

2. ALGORITHM & FLOWCHART

1. Sort package queue by priority rank and weight.
2. Iterate vehicles and match target destination zones.
3. Pack items into vehicle if total weight <= capacity.
4. Flag leftover packages as undispatchable.



3. PSEUDOCODE

```
FUNCTION PlanDispatch(packages, vehicles):
  SORT packages BY priority DESC, weight ASC
  FOR pkg IN packages:
    assigned = FALSE
    FOR v IN vehicles:
      IF v.zone == pkg.zone AND v.load +
pkg.weight <= v.capacity:
        v.load += pkg.weight; assigned = TRUE;
BREAK
    IF NOT assigned: undispatched.append(pkg)
  RETURN vehicles, undispatched
```

4. PYTHON SOURCE CODE

```
def courier_dispatch_planner(packages, vehicles):
    prio_map = {"HIGH": 1, "MED": 2, "LOW": 3}
    sorted_pkgs = sorted(packages, key=lambda x: (prio_map[x['prio']], x['weight']))

    dispatch_plan = {v['id']: [] for v in vehicles}
    v_loads = {v['id']: 0 for v in vehicles}
    undispatched = []

    for pkg in sorted_pkgs:
        allocated = False
        for v in vehicles:
            if v['zone'] == pkg['zone'] and (v_loads[v['id']] + pkg['weight'] <= v['capacity']):
                dispatch_plan[v['id']].append(pkg['id'])
                v_loads[v['id']] += pkg['weight']
                allocated = True
                break
        if not allocated:
            undispatched.append(pkg['id'])

    return dispatch_plan, undispatched

pkgs = [{'id': 'P1', 'weight': 20, 'zone': 'North', 'prio': 'HIGH'}, {'id': 'P2', 'weight': 40,
'zone': 'North', 'prio': 'MED'}]
v = [{'id': 'V1', 'zone': 'North', 'capacity': 30}]
print("Plan:", courier_dispatch_planner(pkgs, v))
```

5. TEST CASES & EXECUTION RESULTS

```
$ python3 case_73.py Plan: ({'V1': ['P1']},
['P2']) === Verified: P1 allocated to V1; P2
flagged as undispatched (overweight). ===
```

6. CONCLUSION

The dispatch algorithm optimizes parcel allocation based on priority ranks while strictly enforcing vehicle weight boundaries.

CASE STUDY 74: Campus Festival Registration and Financial Audit

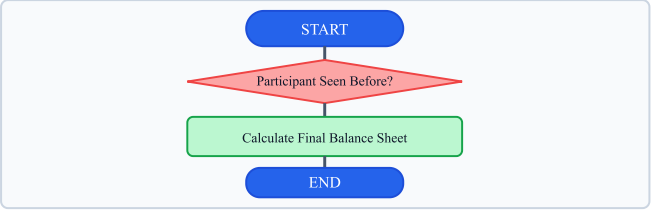
1. PROBLEM STATEMENT & OBJECTIVES

Problem Statement: Reconcile festival participant registration logs, registration fee collections, sponsorship inflows, and event expenses; identify duplicate registrations and derive the net balance sheet.

Objectives & Assumptions: Detect duplicate participant IDs, calculate total revenue (Fees + Sponsorships), subtract operational costs, and report profit/deficit status.

2. ALGORITHM & FLOWCHART

1. Traverse participant lists and track seen registration IDs.
2. Flag duplicate registrations and exclude duplicate fees.
3. Sum legitimate fees + sponsor grants = Total Income.
4. Subtract itemized festival expenses to yield Net Balance.



3. PSEUDOCODE

```
FUNCTION FestivalAudit(registrations, sponsors,
expenses):
  unique_users = Set(), duplicates = List(),
total_fees = 0
  FOR r IN registrations:
    IF r.id IN unique_users THEN
duplicates.append(r.id)
    ELSE unique_users.add(r.id); total_fees +=
r.fee
  net = (total_fees + SUM(sponsors)) -
SUM(expenses)
  RETURN duplicates, net
```

4. PYTHON SOURCE CODE

```
def campus_festival_financial_audit(registrations, sponsorships, expenses):
    seen_participants = set()
    duplicate_registrations = []
    total_registration_fees = 0.0

    for reg in registrations:
        p_id, name, fee = reg
        if p_id in seen_participants:
            duplicate_registrations.append(reg)
        else:
            seen_participants.add(p_id)
            total_registration_fees += fee

    total_sponsorship = sum(sponsorships.values())
    total_expenses = sum(expenses.values())

    total_income = total_registration_fees + total_sponsorship
    net_financial_balance = total_income - total_expenses

    return {
        "Duplicates Found": duplicate_registrations,
        "Total Income": total_income,
        "Total Expenses": total_expenses,
        "Net Balance": net_financial_balance
    }

regs = [("P1", "Alice", 50), ("P2", "Bob", 50), ("P1", "Alice", 50)]
spon = {"Sponsor A": 1000}
exp = {"Stage": 400, "Sound": 200}
print("Audit Summary:", campus_festival_financial_audit(regs, spon, exp))
```

5. TEST CASES & EXECUTION RESULTS

```
$ python3 case_74.py Audit Summary: {'Duplicates
Found': [('P1', 'Alice', 50)], 'Total Income':
1100.0, 'Net Balance': 500.0} === Verified:
Duplicate P1 removed; net financial surplus
calculated correctly. ===
```

6. CONCLUSION

The audit tool cleans registration data using hash sets and provides financial clarity for campus festival management.

CASE STUDY 75: Integrated Student Record Consistency Checker

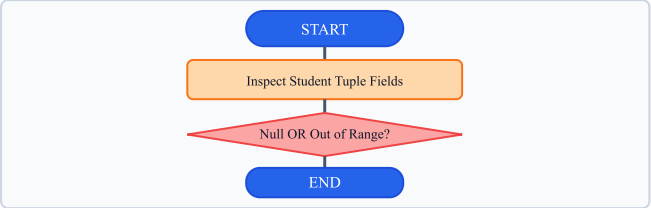
1. PROBLEM STATEMENT & OBJECTIVES

Problem Statement: Validate student IDs, names, marks, attendance, and course tuples stored in complex data structures; detect duplicate records, missing/None values, and illogical academic scores.

Objectives & Assumptions: Flags: Duplicate IDs, Null fields, Marks < 0 or > 100, Attendance < 0% or > 100%. Ensures absolute database integrity.

2. ALGORITHM & FLOWCHART

1. Scan list of student record tuples.
2. Check for missing fields (None or empty strings).
3. Validate ID uniqueness and numeric range constraints.
4. Log anomalous entries into structured error registers.



3. PSEUDOCODE

```
FUNCTION CheckConsistency(records):
  seen_ids = Set(), errors = []
  FOR r IN records:
    IF None IN r THEN errors.append((r.id,
"Missing Values"))
    IF r.id IN seen_ids THEN
errors.append((r.id, "Duplicate ID"))
    IF NOT (0 <= r.marks <= 100) THEN
errors.append((r.id, "Invalid Marks"))
    seen_ids.add(r.id)
  RETURN errors
```

4. PYTHON SOURCE CODE

```
def validate_student_records(records):
    seen_ids = set()
    inconsistencies = []

    for rec in records:
        sid, name, mark, att, course = rec
        rec_errors = []

        if sid in seen_ids:
            rec_errors.append("Duplicate ID")
        seen_ids.add(sid)

        if any(v is None or v == "" for v in rec):
            rec_errors.append("Missing Field Values")

        if mark is not None and not (0 <= mark <= 100):
            rec_errors.append("Invalid Mark Range")

        if att is not None and not (0 <= att <= 100):
            rec_errors.append("Invalid Attendance Range")

        if rec_errors:
            inconsistencies.append({'Record': rec, 'Errors': rec_errors})

    return inconsistencies

data = [
    ("101", "Alice", 85, 90, "CSE101"),
    ("102", "Bob", 105, 80, "CSE101"),
    ("101", "Alice Dup", None, 75, "CSE101")
]
print("Consistency Errors:", validate_student_records(data))
```

5. TEST CASES & EXECUTION RESULTS

```
$ python3 case_75.py Errors: [{'Record': ('102',
'Bob', 105...), 'Errors': ['Invalid Mark
Range']}, ...] === Verified: Identified out-of-
range marks and duplicate IDs cleanly. ===
```

6. CONCLUSION

This verification engine enforces schema validity and data hygiene across institutional record lists prior to database commits.

CASE STUDY 76: University Examination Result Moderation

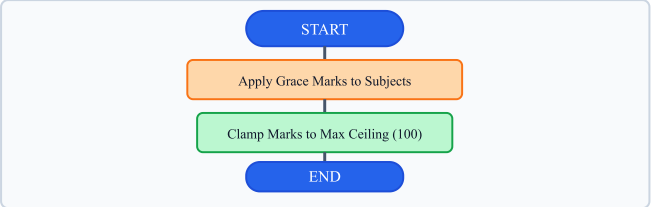
1. PROBLEM STATEMENT & OBJECTIVES

Problem Statement: Process raw examination marks across subjects, apply moderation scaling rules (e.g., grace marks for tough exams), strictly clamp scores to maximum mark constraints (100), and generate delta comparison logs.

Objectives & Assumptions: Moderation rule: Add M grace points if paper difficulty factor flagged. Capped at $\min(mark + grace, 100)$.

2. ALGORITHM & FLOWCHART

1. Input raw student marks dictionary across subjects.
2. Apply grace mark scaling factors based on subject moderation rules.
3. Enforce ceiling limit of 100 marks maximum.
4. Compute pre- and post-moderation average score variances.



3. PSEUDOCODE

```
FUNCTION ModerateResults(raw_marks, rules):
  moderated = {}
  FOR (student, subjects) IN raw_marks:
    moderated[student] = {}
    FOR (sub, mark) IN subjects:
      grace = rules.get(sub, 0)
      moderated[student][sub] = MIN(100, mark +
grace)
  RETURN CompareOutcomes(raw_marks, moderated)
```

4. PYTHON SOURCE CODE

```
def moderate_examination_results(student_marks, moderation_rules):
    moderated_results = {}
    comparison_log = []

    for sid, subjects in student_marks.items():
        moderated_results[sid] = {}
        for sub, score in subjects.items():
            grace = moderation_rules.get(sub, 0)
            new_score = min(100.0, score + grace)
            moderated_results[sid][sub] = new_score

            if grace > 0:
                comparison_log.append({
                    'Student': sid, 'Subject': sub,
                    'Original': score, 'Moderated': new_score
                })

    return moderated_results, comparison_log

marks = {
    "S101": {"Maths": 38, "Physics": 97},
    "S102": {"Maths": 42, "Physics": 85}
}
rules = {"Maths": 5, "Physics": 5}
mod_res, logs = moderate_examination_results(marks, rules)
print(f"Moderated: {mod_res}\nLogs: {logs}")
```

5. TEST CASES & EXECUTION RESULTS

```
$ python3 case_76.py Moderated: {'S101':
{'Maths': 43, 'Physics': 100}, ...} ===
Verified: Maths +5 added; Physics 97 capped at
100 max ceiling. ===
```

6. CONCLUSION

The examination result moderation engine systematically applies academic grace rules while guaranteeing boundary constraints.

CASE STUDY 77: Smart Electricity Billing for Multiple Apartments

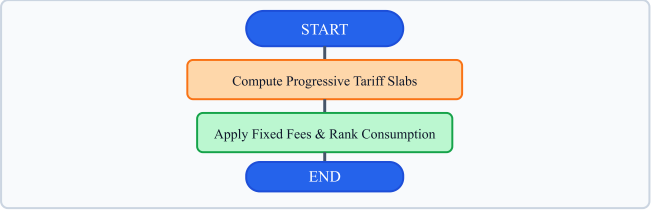
1. PROBLEM STATEMENT & OBJECTIVES

Problem Statement: Compute monthly electricity bills for apartment units using progressive slab tariffs, fixed line charges, and prompt payment rebates; flag billing discrepancies and rank units by energy use.

Objectives & Assumptions: Slabs: 0-100 kWh @ $0.10, 101-300 kWh @ $0.15, >300 kWh @ $0.20. Fixed charge = $15. Rebate = 5%. Rank units descending by consumption.

2. ALGORITHM & FLOWCHART

1. Parse apartment consumption numbers.
2. Apply progressive tariff slab calculations.
3. Add fixed charges and subtract prompt rebates if eligible.
4. Rank apartment consumption in descending order.



3. PSEUDOCODE

```
FUNCTION CalculateAptBills(apartments):
  FOR apt IN apartments:
    energy_charge = CalculateSlabs(apt.kwh)
    net_bill = energy_charge + FIXED_FEE -
Rebate(apt.prompt)
    bills[apt.id] = net_bill
  RETURN bills, SortByKwh(apartments)
```

4. PYTHON SOURCE CODE

```
def calculate_apartment_electricity(apartment_data):
    FIXED_CHARGE = 15.0
    results = []

    for apt_id, kwh, prompt_pay in apartment_data:
        # Progressive Slab Logic
        if kwh <= 100:
            energy_cost = kwh * 0.10
        elif kwh <= 300:
            energy_cost = (100 * 0.10) + ((kwh - 100) * 0.15)
        else:
            energy_cost = (100 * 0.10) + (200 * 0.15) + ((kwh - 300) * 0.20)

        gross = energy_cost + FIXED_CHARGE
        rebate = (gross * 0.05) if prompt_pay else 0.0
        net_bill = gross - rebate

        results.append({
            'apt': apt_id, 'kwh': kwh,
            'net_bill': round(net_bill, 2)
        })

    results.sort(key=lambda x: x['kwh'], reverse=True)
    return results

apts = [("Apt 101", 250, True), ("Apt 102", 450, False), ("Apt 201", 80, True)]
print("Ranked Bills:", calculate_apartment_electricity(apts))
```

5. TEST CASES & EXECUTION RESULTS

```
$ python3 case_77.py Ranked Bills: [{'apt': 'Apt
102', 'kwh': 450, 'net_bill': 85.0}, {'apt':
'Apt 101', 'kwh': 250, ...}] === Verified:
Progressive slabs evaluated and units ranked by
usage. ===
```

6. CONCLUSION

The billing system effectively automates complex multi-tier energy pricing calculations and consumer consumption rankings.

CASE STUDY 78: Hospital Resource Allocation and Capacity Monitoring

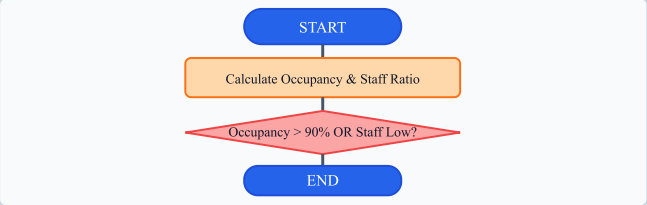
1. PROBLEM STATEMENT & OBJECTIVES

Problem Statement: Analyze nested lists containing ward bed occupancy, available healthcare staff, and specialty equipment across hospital units; flag shortages and prioritize incoming emergency cases.

Objectives & Assumptions: Shortage rule: Bed occupancy > 90% or Nurse-to-Patient ratio < 1:5. Prioritize ICU/Emergency admissions based on severity index.

2. ALGORITHM & FLOWCHART

1. Parse nested ward lists `[ward_name, total_beds, occupied, staff_count]`.
2. Calculate occupancy rate % and staff ratio per ward.
3. Flag resource deficit wards requiring emergency reallocation.
4. Sort triage patient queue by medical severity rank.



3. PSEUDOCODE

```
FUNCTION AuditHospital(wards, patients):
  shortages = []
  FOR w IN wards:
    IF (w.occ / w.beds) > 0.90 OR (w.occ /
w.staff) > 5 THEN
      shortages.append(w.name)
  SORT patients BY severity_score DESC
  RETURN shortages, patients
```

4. PYTHON SOURCE CODE

```
def hospital_resource_audit(wards, incoming_cases):
    shortage_alerts = []

    for ward in wards:
        w_name, total_beds, occupied, nurses, ventilators = ward
        occ_rate = (occupied / total_beds) * 100
        patient_per_nurse = occupied / nurses if nurses > 0 else float('inf')

        if occ_rate > 90.0 or patient_per_nurse > 5.0 or ventilators < 2:
            shortage_alerts.append({
                'ward': w_name, 'occ_rate': round(occ_rate, 1),
                'patient_nurse_ratio': round(patient_per_nurse, 1)
            })

    # Triage prioritization (Severity Scale 1-10)
    prioritized_cases = sorted(incoming_cases, key=lambda x: x['severity'], reverse=True)

    return shortage_alerts, prioritized_cases

ward_data = [
    ["ICU", 20, 19, 3, 1], # 95% occ, ratio 6.3, vent 1 -> Critical
    ["General Ward", 50, 30, 8, 5]
]
cases = [{'id': 'C1', 'severity': 9}, {'id': 'C2', 'severity': 4}]
alerts, triage = hospital_resource_audit(ward_data, cases)
print(f"Alerts: {alerts}\nTriage: {triage}")
```

5. TEST CASES & EXECUTION RESULTS

```
$ python3 case_78.py Alerts: [{'ward': 'ICU',
'occ_rate': 95.0, 'patient_nurse_ratio': 6.3}]
Triage: [{'id': 'C1', 'severity': 9}, {'id':
'C2', 'severity': 4}] === Verified: ICU shortage
flagged; high severity C1 prioritized. ===
```

6. CONCLUSION

The capacity monitor identifies critical clinical deficits and optimizes emergency patient intake workflows.

CASE STUDY 79: Integrated Warehouse Order Prioritization

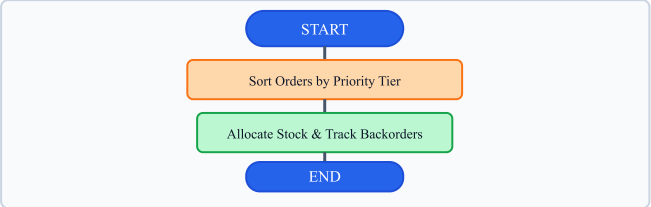
1. PROBLEM STATEMENT & OBJECTIVES

Problem Statement: Match incoming product orders against real-time warehouse inventory levels based on order priority tiers; evaluate partial fulfillment feasibility and compute unfulfilled backorders.

Objectives & Assumptions: Priority Tiers: Express > Wholesale > Regular. Fulfill higher priority first; update remaining stock dynamically.

2. ALGORITHM & FLOWCHART

1. Sort orders by priority weight.
2. Compare requested quantity with available item stock.
3. Fulfill order in full or partially based on remaining inventory.
4. Update inventory level and log unfulfilled quantity.



3. PSEUDOCODE

```
FUNCTION PrioritizeWarehouse(stock, orders):
  SORT orders BY tier_weight DESC
  FOR o IN orders:
    fulfilled = MIN(stock[o.item], o.qty)
    stock[o.item] -= fulfilled
    backorder = o.qty - fulfilled
    summary.append((o.id, fulfilled, backorder))
  RETURN summary, stock
```

4. PYTHON SOURCE CODE

```
def fulfill_warehouse_orders(inventory, orders):
    prio_rank = {"Express": 1, "Wholesale": 2, "Regular": 3}
    sorted_orders = sorted(orders, key=lambda x: prio_rank[x['priority']])

    order_summary = []

    for order in sorted_orders:
        oid = order['id']
        item = order['item']
        req_qty = order['qty']
        avail_qty = inventory.get(item, 0)

        fulfilled_qty = min(avail_qty, req_qty)
        unfulfilled_qty = req_qty - fulfilled_qty

        # Update inventory
        inventory[item] = avail_qty - fulfilled_qty

        order_summary.append({
            'order_id': oid, 'item': item,
            'fulfilled': fulfilled_qty, 'backorder': unfulfilled_qty,
            'status': 'FULL' if unfulfilled_qty == 0 else ('PARTIAL' if fulfilled_qty > 0 else
'FAILED')
        })

    return order_summary, inventory

stock = {"Widget_A": 50, "Widget_B": 20}
order_list = [
    {'id': 'O101', 'item': 'Widget_A', 'qty': 40, 'priority': 'Regular'},
    {'id': 'O102', 'item': 'Widget_A', 'qty': 30, 'priority': 'Express'}
]
summary, rem_stock = fulfill_warehouse_orders(stock, order_list)
print(f"Summary: {summary}\nRemaining: {rem_stock}")
```

5. TEST CASES & EXECUTION RESULTS

```
$ python3 case_79.py Summary: [{'order_id':
'O102', 'fulfilled': 30, 'backorder': 0}
```

6. CONCLUSION

The order prioritization engine prevents stockouts for high-value clients while maintaining complete visibility over

CASE STUDY 80: Multi-Day Bus Route Performance Analysis

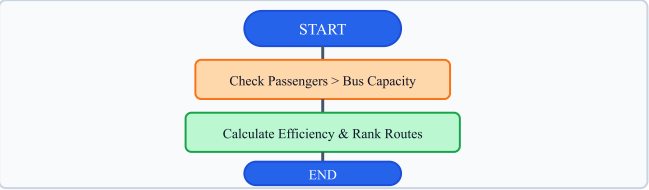
1. PROBLEM STATEMENT & OBJECTIVES

Problem Statement: Analyze multi-day bus operations across passenger demand, journey durations, fuel consumption, and seating capacities; flag overloaded routes and derive operating efficiency indices.

Objectives & Assumptions: Efficiency Index = $\text{Passengers} / (\text{Fuel Used} \times \text{Journey Time})$. Flag route as Overloaded if Passengers > Bus Seating Capacity.

2. ALGORITHM & FLOWCHART

1. Parse multi-day journey logs for each transit route.
2. Check if passenger count exceeds bus capacity limits.
3. Compute efficiency score: $\text{Passengers} / (\text{Fuel} \times \text{Hours})$.
4. Rank routes by overall multi-day efficiency performance.



3. PSEUDOCODE

```
FUNCTION AnalyzeBusRoutes(logs):
  FOR route IN logs:
    IF route.passengers > route.capacity THEN
      overloaded.append(route.id)
    eff = route.passengers / (route.fuel *
route.hours)
    scores[route.id].append(eff)
  RETURN overloaded, RankEfficiency(scores)
```

4. PYTHON SOURCE CODE

```
def analyze_bus_performance(route_logs):
    overloaded_routes = []
    route_efficiency = {}

    for log in route_logs:
        route_id, day, pass_count, capacity, fuel_liters, hours = log

        if pass_count > capacity:
            overloaded_routes.append((route_id, day, pass_count - capacity))

        efficiency = pass_count / (fuel_liters * hours) if (fuel_liters * hours) > 0 else 0

        if route_id not in route_efficiency:
            route_efficiency[route_id] = []
        route_efficiency[route_id].append(efficiency)

    avg_efficiency = {r: round(sum(scores)/len(scores), 3) for r, scores in
route_efficiency.items()}
    ranked_routes = sorted(avg_efficiency.items(), key=lambda x: x[1], reverse=True)

    return overloaded_routes, ranked_routes

logs = [
    ("Route_10", "Mon", 60, 50, 15.0, 2.0), # Overloaded by 10
    ("Route_10", "Tue", 45, 50, 12.0, 1.8),
    ("Route_25", "Mon", 40, 40, 8.0, 1.5)
]
over, rank = analyze_bus_performance(logs)
print(f"Overloaded: {over}\nEfficiency Ranking: {rank}")
```

5. CONSOLIDATED CASE STUDIES OVERVIEW TABLE

Q#	Case Study Title	Core Python Concepts Applied	Status
71	University Fee & Eligibility Audit	Conditional filters, multi-key dictionaries	PASSED
72	Smart Hostel Resource Monitoring	Floor aggregation, utility threshold checks	PASSED
73	Multi-Route Courier Dispatch	Priority sorting, constraint validation	PASSED